Unit #: 2

Unit Name: Counting, Sorting and Searching

Topic: Implementations of Matrix operations

Course objectives:

The objective(s) of this course is to make students

CObj1: Acquire knowledge on how to solve relevant and logical problems using computing machine

CObj2: Map algorithmic solutions to relevant features of C programming language constructs

CObj3: Gain knowledge about C constructs and its associated eco-system

CObj4: Appreciate and gain knowledge about the issues with C Standards and its respective behaviors

CObj5: Get insights about testing and debugging C Programs

Course outcomes:

At the end of the course, the student will be able to

CO1: Understand and apply algorithmic solutions to counting problems using appropriate C Constructs

CO2: Understand, analyse and apply text processing and string manipulation methods using C Arrays, Pointers and functions

CO3: Understand prioritized scheduling and implement the same using C structures

CO4: Understand and apply sorting techniques using advanced C constructs

CO5: Understand and evaluate portable programming techniques using preprocessor directives and conditional compilation of C Programs

Let us write separate functions to **add, subtract and multiply two matrices**. Display appropriate message when two matrices are incompatible for these operations. Add the implementations of other operations such as finding the transpose of a matrix, identity matrix or not etc

Coding Example_1:

```
void read(int (*a1)[],int,int);
void display( int (*a1)[],int,int);
void multiply(int n1; int n2; int (*a)[n1],int (*b)[n2],int m1,int n1,int n2,int (*c)[n2]);
void subtract(int n; int (*a)[n],int (*b)[n],int m, int n, int (*c)[n]);
void add(int n; int (*a)[n],int (*b)[n],int m, int n, int (*c)[n]);
int main()
{
	int a[100][100];
	int b[100][100];
	int c[100][100];
	int m1,n1;
	int m2,n2;
	printf("enter the order of a\n");
	scanf("%d%d",&m1,&n1);
	printf("enter the elements of a\n");
	read(a,m1,n1);
	printf("enter the order of b\n");
	scanf("%d%d",&m2,&n2);
	printf("enter the elements of b\n");
	read(b,m2,n2);
	if (m1 == m2 && n1 == n2)
	{
		printf("elements of A are\n");
		display(a,m1,n1);
		printf("elements of B are\n");
```

```
        display(b,m2,n2);
        printf("Addition of two matrices\n");
        add(a,b,m1,n1,c);
        display(c,m2,n2);
        printf("Subtraction of two matrices\n");
        subtract(a,b,m1,n1,c);
        display(c,m2,n2);

    }
    else

    {   printf("Two matrices are not compatible for addition and subtraction\n");     }
    if (n1==m2)
    {       printf("Multiplication of two matrices\n");
        multiply(a,b,m1,n1,n2,c);
        display(c,m1,n2);
    }
    else
    {     printf("Two matrices are not compatible for multiplication too\n");       }
    return 0;
}

void add(int n; int (*a)[n],int (*b)[n],int m, int n, int (*c)[n])
{
    for(int i = 0;i<m;i++)
    {
        for(int j= 0;j<n;j++)
        {
            c[i][j] = a[i][j]+b[i][j];
        }
    }
}
```

```
void subtract(int n; int (*a)[n],int (*b)[n],int m, int n, int (*c)[n])
{
	for(int i = 0;i<m;i++)
	{
		for(int j= 0;j<n;j++)
			{
				c[i][j] = a[i][j] - b[i][j];
			}
	}
}
void multiply(int n1; int n2; int (*a)[n1],int (*b)[n2],int m1,int n1,int n2,int (*c)[n2])
{
	for(int i = 0;i<m1;i++)
	{
		for(int j= 0;j<n2;j++)
		{
			c[i][j] = 0;
		}
	}
	for(int i=0;i<m1;i++)
	{
		for(int j=0;j<n2;j++)
		{
			int sum=0;
			for(int k=0;k<n1;k++)
			{	sum=sum+a[i][k]*b[k][j];	}
			c[i][j]=sum;
		}
	}
}
```

Think about sending only a and b matrices to add, subtract and multiply functions and c is the result of operation and is the returned value from these functions. Is it correct to implement this way?

Coding Example_2: Let us write a program to take n names from the user and print it. Each name can have maximum of 20 characters.

```
#include<stdio.h>
 int main()
 {
	// Think about the commented code? Is it correct?
	/*
	char name1[20]; //Can store one name with maximum of 20 characters including '\0'
	// So need n different variables to store n names
	// But can we store n names under the same variable name???? --- Yes. Use 2D array
	*/
	char names[200][20];
	int n;
	printf("How many names you want to store?\n");
	scanf("%d", &n);
	int i;
	printf("enter %d names\n", n); for(i=0;i<n;i++)
	{
		scanf("%[^\n]s",names[i]);
	}
	printf("U entered below names\n"); for(int i=0;i<n;i++)
	{
		printf("%s\n",names[i]);
	}
	return 0;
}
```

Happy Coding!